SC 617 - Adaptive Control

Homework

March 22, 2026

Date Assigned: March 22, 2026
Date Due: March 31, 2026

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1 Basic Equations of Rigid Body Attitude Dynamics

The dynamics of rotation can be described in terms of the rotation matrix or the corresponding quarternion. In terms of the rotation matrix $\mathbf{R}$, the dynamics are given by,

$$\dot{\mathbf{R}} = -\mathbf{S}(\boldsymbol{\omega})\mathbf{R} \tag{1}$$

where $\boldsymbol{\omega}$ is the angular velocity and $\mathbf{S}(\boldsymbol{\omega})$ is defined by $\mathbf{S}(\boldsymbol{a})\boldsymbol{b} = \boldsymbol{a} \times \boldsymbol{b}$.
In terms of the quarternion $\boldsymbol{q} = \left[q_o \ \boldsymbol{q}_v^T\right]^T$, the dynamics are given by

$$\begin{pmatrix} \dot{q}_o \\ \dot{\boldsymbol{q}}_v \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -\boldsymbol{q}_v^T \\ q_o\mathbf{I} + \mathbf{S}(\boldsymbol{q}_v) \end{pmatrix}\boldsymbol{\omega} := \frac{1}{2}E(\boldsymbol{q})\boldsymbol{\omega} \tag{2}$$

The angular velocity dynamics are given by the equations

$$\mathbf{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} + \boldsymbol{u} \tag{3}$$

where $\mathbf{J} \in \mathbb{R}^{3\times 3}$ ($\mathbf{J} = \mathbf{J}^T > 0$) is the positive-definite *inertia matrix* and $\boldsymbol{u}$ is external input (eg. actuation from thrusters) to the system. In most cases $\mathbf{J}$ will not be known accurately and thus the control algorithms may require adaptation on $\mathbf{J}$.

2 Tracking Problem

For the tracking problem, we want $\boldsymbol{q}$ to follow the desired trajectory $\boldsymbol{q}_d(t)$ and $\boldsymbol{\omega}$ to follow the desired trajectory $\boldsymbol{\omega}_d(t)$.

- $\mathbf{R}(\boldsymbol{q})$ denotes the rotation matrix corresponding to the quarternion $\boldsymbol{q}$.
- $\mathbf{R}_d(t) := \mathbf{R}(\boldsymbol{q_d}(t))$ denotes the desired orientation trajectory in terms of the rotation matrix corresponding to quaternion $\boldsymbol{q_d}$.
- The desired angular velocity $\boldsymbol{\omega}_d(t)$ is usually expressed in the coordinates corresponding to the desired orientation $\mathbf{R}_d(t)$.

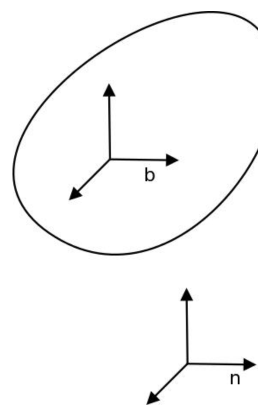

Figure 1: Inertial (n) and body (b) coordinates

$$\begin{aligned} \mathcal{B} \xrightarrow{\mathbf{R}(\boldsymbol{q})} \mathcal{N} \qquad & \{\hat{\boldsymbol{b}}\} = \mathbf{R}(\boldsymbol{q})\{\hat{\boldsymbol{n}}\} \\ \mathcal{D} \xrightarrow{\mathbf{R}(\boldsymbol{q}_d)} \mathcal{N} \qquad & \{\hat{\boldsymbol{d}}\} = \mathbf{R}(\boldsymbol{q}_d)\{\hat{\boldsymbol{n}}\} \\ \mathcal{B} \xrightarrow{\mathbf{R}(\boldsymbol{s})} \mathcal{D} \qquad & \{\hat{\boldsymbol{b}}\} = \mathbf{R}(\boldsymbol{s})\{\hat{\boldsymbol{d}}\} \end{aligned}$$

where $\mathcal{B}$: body frame coordinates, $\mathcal{N}$: inertial frame coordinates, $\mathcal{D}$: desired body frame coordinates. The quantities in the curly braces represent the unit vectors in the corresponding frames.

$\mathbf{R}(\boldsymbol{s}) \in \mathcal{SO}(3)$ represents the orientation error since it gives the difference between the desired orientation and the current body frame orientation. Thus, $\boldsymbol{s}$ gives the *error quarternion.* From the above transformations we get

$$\mathbf{R}(\boldsymbol{q})\{\hat{\boldsymbol{n}}\} = \mathbf{R}(\boldsymbol{s})\mathbf{R}(\boldsymbol{q}_d)\{\hat{\boldsymbol{n}}\} \tag{4}$$

$$\implies \mathbf{R}(\boldsymbol{s}) = \mathbf{R}(\boldsymbol{q})\mathbf{R}(\boldsymbol{q}_d)^T \tag{5}$$

When $\boldsymbol{q} = \boldsymbol{q}_d$,

$$\boldsymbol{s} = \begin{pmatrix} \pm 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

The matching condition is given by

$$\dot{\mathbf{R}}_d = -\mathbf{s}(\boldsymbol{\omega_d})\mathbf{R}_d \tag{6}$$

3 Error Dynamics

Taking the derivative of equation (5),

$$\dot{\mathbf{R}}_s = \dot{\mathbf{R}}\mathbf{R}_d^T + \mathbf{R}\dot{\mathbf{R}}_d^T \tag{7}$$

where we use the convenient notation $\mathbf{R}_s := \mathbf{R}(\boldsymbol{s})$, $\mathbf{R} := \mathbf{R}(\boldsymbol{q})$ and $\mathbf{R}_d := \mathbf{R}(\boldsymbol{q}_d)$.
Now,

$$\begin{aligned} &\frac{d}{dt}(\mathbf{R}_d\mathbf{R}_d^T) = 0 \quad \text{since} \quad \mathbf{R}_d\mathbf{R}_d^T = \mathbf{I} \\ \implies\ & \dot{\mathbf{R}}_d\mathbf{R}_d^T + \mathbf{R}_d\dot{\mathbf{R}}_d^T = 0 \\ \implies\ & \dot{\mathbf{R}}_d^T = -\mathbf{R}_d^T\dot{\mathbf{R}}_d\mathbf{R}_d^T \end{aligned}$$

Using the above equation and equation (1) in equation (7), we get

$$\begin{aligned} \dot{\mathbf{R}}_s &= -\mathbf{S}(\boldsymbol{\omega})\mathbf{R}\mathbf{R}_d^T + \mathbf{R}\mathbf{R}_d^T\mathbf{S}(\boldsymbol{\omega}_d)\mathbf{R}_d\mathbf{R}_d^T \\ &= -\mathbf{S}(\boldsymbol{\omega})\mathbf{R}_s + \mathbf{R}_s\mathbf{S}(\boldsymbol{\omega}_d) \\ \dot{\mathbf{R}}_s\boldsymbol{x} &= -\boldsymbol{\omega} \times (\mathbf{R}_s\boldsymbol{x}) + \mathbf{R}_s(\boldsymbol{\omega}_d \times \boldsymbol{x}) \\ &= -\boldsymbol{\omega} \times (\mathbf{R}_s\boldsymbol{x}) + (\mathbf{R}_s\boldsymbol{\omega}_d) \times (\mathbf{R}_s\boldsymbol{x}) \\ &= -(\boldsymbol{\omega} - \mathbf{R}_s\boldsymbol{\omega}_d) \times \mathbf{R}_s\boldsymbol{x} \end{aligned}$$

$$\implies \dot{\mathbf{R}}_s = -\mathbf{S}(\boldsymbol{\omega} - \mathbf{R}_s\boldsymbol{\omega}_d)\mathbf{R}_s \tag{8}$$

$\mathbf{R}_s\boldsymbol{\omega}_d$ represents $\boldsymbol{\omega}_d$ expressed in the body frame. It can be shown from equation (8) that

$$\dot{\boldsymbol{s}} = \frac{1}{2}\mathbf{E}(\boldsymbol{s})(\boldsymbol{\omega} - \mathbf{R}_s\boldsymbol{\omega}_d) \tag{9}$$

Next we define the error in $\boldsymbol{\omega}$ as $\delta\boldsymbol{\omega} := \boldsymbol{\omega} - \mathbf{R}_s\boldsymbol{\omega}_d$. Then,

$$\mathbf{J}\dot{\delta\boldsymbol{\omega}} = \mathbf{J}\dot{\boldsymbol{\omega}} - \mathbf{J}(\dot{\mathbf{R}}_s\boldsymbol{\omega}_d + \mathbf{R}_s\dot{\boldsymbol{\omega}}_d)$$

$$\implies \mathbf{J}\dot{\delta\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} + \boldsymbol{u} - \mathbf{J}(\dot{\mathbf{R}}_s\boldsymbol{\omega}_d + \mathbf{R}_s\dot{\boldsymbol{\omega}}_d) \tag{10}$$

Equations (9) and (10) gives the *error dynamics* for the tracking problem.

4 Tracking in case of known J

The tracking objectives in terms of the error dynamics are $\boldsymbol{s}_v \to 0$, $\delta\boldsymbol{\omega} \to 0$.
Choose the candidate Lyapunov function,

$$V = \frac{1}{2}\delta\boldsymbol{\omega}^T \mathbf{J}\delta\boldsymbol{\omega} + k_p\left[\boldsymbol{s}_v^T\boldsymbol{s}_v + (1 - s_o)^2\right] \tag{11}$$

Then,

$$\begin{aligned}\dot{V} &= \delta\boldsymbol{\omega}^T\left[-\boldsymbol{\omega}\times\mathbf{J}\boldsymbol{\omega} + \boldsymbol{u} - \mathbf{J}\boldsymbol{\phi}\right] - 2k_p\dot{s}_o \\ &= \delta\boldsymbol{\omega}^T\left[-\boldsymbol{\omega}\times\mathbf{J}\boldsymbol{\omega} + \boldsymbol{u} - \mathbf{J}\boldsymbol{\phi}\right] + k_p\boldsymbol{s}_v^T\delta\boldsymbol{\omega}\end{aligned}$$

where $\boldsymbol{\phi} := \dot{\mathbf{R}}_s\boldsymbol{\omega}_d + \mathbf{R}_s\dot{\boldsymbol{\omega}}_d$. Notice that $\mathbf{J}$ is symmetric and hence has only 6 distinct entries, which we denote as $\boldsymbol{\theta}^*$. Further we can define, $W\boldsymbol{\theta}^* := -\boldsymbol{\omega}\times\mathbf{J}\boldsymbol{\omega} - \mathbf{J}\boldsymbol{\phi}$ for some $W(\cdot) \in \mathbb{R}^{3\times 6}$ since the terms are linear in $\mathbf{J}$. Now we choose

$$\boldsymbol{u} = -k_p\boldsymbol{s}_v - k_w\delta\boldsymbol{\omega} - W\boldsymbol{\theta}^* \tag{12}$$

This yields

$$\dot{V} = -k_w||\delta\boldsymbol{\omega}||^2 \leq 0 \tag{13}$$

Thus $\dot{V}$ is negative semi-definite and using signal chasing analysis and Barbalat's lemma, it can be shown that

$$\begin{aligned} & \delta\boldsymbol{\omega} \to 0 \\ & \delta\dot{\boldsymbol{\omega}} \to 0 \\ \implies\ & \boldsymbol{s}_v \to 0 \qquad \text{(closed loop dynamics is } \mathbf{J}\delta\dot{\boldsymbol{\omega}} = -k_p\boldsymbol{s}_v - k_w\delta\boldsymbol{\omega}) \end{aligned}$$

Thus the tracking objective is achieved with the control law $\boldsymbol{u}$ given by equation (12).

5 Spacecraft Data

Let the inertia matrix for a rigid spacecraft br described by the matrix,

$$J = \begin{bmatrix} 20 & 1.2 & 0.9 \\ 1.2 & 17 & 1.4 \\ 0.9 & 1.4 & 15 \end{bmatrix}$$

Assuming the spacecraft is initially at rest and oriented in a direction represented by the quaternion,

$$\boldsymbol{q}_v(0) = [0.1826\ 0.1826\ 0.1826]^T \quad q_o(0) = -\sqrt{1 - \|\boldsymbol{q}_v(0)\|^2}$$

6 Homework Tasks

You are required to perform the following activities,

(a) Simulate the motion of the spacecraft for $t \in [0, 60]$ sec in the absence of any control assuming any non zero initial condition on ω. Plot $\omega(t)$, $\|\boldsymbol{q}_v\|$ and q_o with time. Do the trajectories remain bounded ? Verify that the unit norm constraint on the quaternions is maintained. (10)

(b) Design and simulate a control law $u(t)$ for the spacecraft to achieve stabilization. Plot the state and the control signals. The gain parameters in your design must be chosen to ensure that the norm of the state vector $[\omega,\ \boldsymbol{q}_v]^T$ remains less than 10^{-3} for all $t \geq 20$. (10)

(c) Consider the trajectory tracking problem for the aforementioned spacecraft. The desired angular velocity trajectory in a desired reference frame is given by $\omega_d(t) = [0.3\cos(t)(1 - e^{-0.01t^2}) + (0.08\pi + 0.006\sin(t))te^{-0.01t^2}, 0.3\cos(t)(1 - e^{-0.01t^2}) + (0.08\pi + 0.006\sin(t))te^{-0.01t^2}, 1]^\top$. The desired reference frame $\boldsymbol{q}_d(t)$ is obtained from $\omega_d(t)$ by solving the corresponding quaternion dynamics ($\dot{\boldsymbol{q}}_d = 0.5\ E(\boldsymbol{q}_d)\omega_d$) with the following initial conditions $\boldsymbol{q}_{d,o}(0) = 1 \quad \boldsymbol{q}_{d,v}(0) = [0\ 0\ 0]^T$. Assuming that the inertia matrix is perfectly defined, design and simulate a full state feedback attitude controller that tracks the desired trajectory. Plot the error quaternion, angular velocity and the control. Choose the gains so as to make sure that magnitude of control required on each axis is less than 5 Nm. (15)

(d) Consider the spacecraft tracking error dynamics (9)-(10). Derive a control law $\boldsymbol{u}(t)$ to guarantee perfect attitude and angular velocity tracking using an integrator backstepping type construction proposed for linear double integrators for the known inertia case. Provide complete stability analysis along with the control design. (20)

(e) Design and simulate an adaptive attitude tracking controller for the above case when the inertia matrix is unknown but a symmetric positive definite constant matrix. Assume starting inertial values at a 30% off-set from the true value. Provide plots of error quaternion, angular velocity, control and the parameter error. Do the parameters converge to their true values ? Bonus points if you can alter desired trajectory to ensure convergence. (15)

(f) Consider the alternate form of (9)-(10) below,

$$\dot{\boldsymbol{s}} = \frac{1}{2}\mathbf{E}(\boldsymbol{s})\delta\boldsymbol{\omega}$$
$$\mathbf{J}\dot{\delta\boldsymbol{\omega}} = -\boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} + \boldsymbol{\nu} - \mathbf{J}\boldsymbol{\phi}$$
$$\dot{\boldsymbol{\nu}} = u$$

where $\boldsymbol{\nu} \in \mathbb{R}^3$ is the control. Design an adaptive backstepping ? Provide simulations with same data as in part (c) and assuming starting inertial values at a 30% off-set from the true value. (30)

References

[1] M. Krstic, I. Kanellakopoulos, and P. V. Kokotovic, *Nonlinear and Adaptive Control Design*, 1st ed., ser. Adaptive and Learning Systems for Signal Processing, Communications and Control Series. Wiley-Interscience, 1995.